



COURSE SYLLABUS					
Cours e Code	COMP 030	Course Title	Business Intelligence		Course Credit 3
Course Description		Companies recognize the importance of corporate data and information to increase competitive advantage, thus business intelligence is used. Business intelligence (BI) is a broad category of applications, technologies, and processes for gathering, storing, accessing, and analyzing data to help business users make better decisions and take actions. It is about how organizations can successfully "collect, evaluate and apply information" to become better decision makers. It starts with basic concepts regarding business needs, and ends with hands-on experience using Business Intelligence tools. The development and use of data warehouses and data marts, and the application of selected data mining techniques to business decision making is illustrated. Active participation of students in the delivery of the course through case and project presentation is a requirement.			
Pre-Requisites		None	Co-Requisite s	None	
VISION		A Leading Comprehensive Polytechnic University in Asia			
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.			
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.			



INSTITUTIONAL
LEARNING OUTCOMES
(ILO)

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.																		
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.																		
PROGRAM LEARNING OUTCOMES	Based on CMO Alignment to ILOs <table><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	1	2	3	4	5	6	7	8	9									
1	2	3	4	5	6	7	8	9											



(PLO)	Apply knowledge of computing, science, and mathematics appropriate to the discipline.														
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution														
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.														
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.														



	Integrate IT-based solutions into the user environment effectively.												
	Assist in the creation of an effective IT project plan.												
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession												
	Understand best practices and standards and their applications.												
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.												
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.												
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.												
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.												
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development												

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Use Business Intelligence systems and technology to support decision making.								
	Design and build Business Intelligence applications based on users' needs.								



	Identify business and technical requirements for a BI solution.								
	Apply relevant theories, concepts and techniques to solve real-world BI problems.								
	Perform data analyses.								
	Visualize the results of data analyses.								
	Articulate examples of how businesses are using BI to enhance competitiveness and profitability. Evaluate the limitations and possibilities of BI technology.								
	Recommend BI tools and practices that align with business strategies based on case analyses.								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content / Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	Identify the requirements and expectations from the course	Course Orientation	Active/ Collaborative participation in Lectures, Discussion Forums, Workshops, Laboratory Exercises, Individual activities, and group projects.		Seminars, Assignment, Laboratory Exercises, Projects, Quizzes, Midterm & Final					
2	Familiarize with BI, Business Analytics, and Decision Support Systems terminology, concepts, and issues	Introduction to Business Intelligence								



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3	Compare operational databases and data warehouses Discuss data marts and warehouse architectures	Data Warehousing								
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4	Explain the ETL (Extraction, Transformation, Loading) process Articulate data warehousing development/impl ementation and administration issues Analyze future data warehousing trends	Business Analytics and Data Visualization								
5-6	Describe the concept of 'data mining' Analyze different data types and data mining application domains Describe the process of data mining	Business Driven Data Mining Objectives and Process								



7-8	Apply classification techniques Apply hierarchical clustering techniques Mine association rules from given data	Data Mining Techniques								
9	MIDTERM EXAMINATION									
10-11	Discuss Natural Language Processing Apply text mining to a collection of documents Apply sentiment analysis to user-generated content Describe web mining and social media analytics	Text and Web Analytics								



12	Identify approaches to business reporting Apply visual analytics	Business Reporting and Visual Analytics								
13	Elaborate Social Network Analysis (SNA) Apply SNA to network/online communication data Compare community detection approaches	Social Network Analysis and Community Detection								



14	Analyze traditional vs. 'big' data collections Apply Hadoop and NoSQL Discover major big data vendors	Introduction to Big Data								
15	Apply big data and stream analytics	Big Data Analytics								



16	Explain location-based and consumer analytics Analyze recommendation engines Discuss BI/BA in cloud computing Evaluate legality, privacy, and ethics in the BI/BA context	Emerging BI/BA Trends and Future Impacts									
17	FINAL EXAMINATION										
17		Project / Case Presentation									



18		Round-up Activities									
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LIST OF REFERENCES

1. Turban, Efraim, Shadra, Ramesh & Delen, Dursun. Decision Support and Business Intelligence Systems: International Edition, 9Th edition. Pearson Publishing, 2011.
2. Galit Shmueli, Nitin Patel, and Peter Bruce. Data Mining for Business Intelligence: Concepts, Technologies, and Applications in Micorsoft Office Excel with XLminer, John Wiley & Sons, Inc. 2006.
3. Michael J. A. Berry and Gordon S. Linoff, Mastering Data Mining: The Art and Science of Customer Relationship Management, Wiley, 2000.
4. Groth, Robert, Data Mining: Building Competitive Advantage, Prentice Hall, 2000.
5. Kudyba, Stephan, & Hoptroff, Richard, Data Mining and Business Intelligence: A Guide to Productivity,
6. Sharda, Ramesh, Business Intelligence: A Managerial Perspectife on Analytics 3rd Ed.
7. Scheps. Swaine, Business Intelligence for Dummies.

WEBSITE:

<http://www.businessintelligence.com/>
<http://www.sas.com/technologies/bi/>
<http://www.microsoft.com/bi/>
<http://www.dwinfocenter.org/>
<http://www.data-mining-guide.net/>
<http://www.adastracorp.com/>
<http://www.kdnuggets.com/datasets/>
<http://www.dataminingbook.com/>
http://www.kmining.com/info_conferences.html
<http://www.resample.com/xlminer/help/Index.htm>



Rubrics

% out of maximum	Definition	Criteria
85-10	Excellent	An excellent result in terms of theoretical depth, practical relevance, analytical ability and independence.
75-84	Very Good	A very good result in terms of theoretical depth, practical relevance, analytical ability and independence.
65-74	Good	A good result in terms of theoretical depth, practical relevance, analytical ability and independence.
55-64	Satisfactory	A satisfactory result in terms of theoretical depth, practical relevance, analytical ability and independence.
50-54	Sufficient	A result that satisfies the minimum requirements wit regard to theoretical depth, practical relevance, analytical ability and independence.
0-49	Fail	An inadequate result with regard to theoretical depth, practical relevance, analytical ability and independence.



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General Rules

1. The course is expected to have a minimum of three (3) quizzes. **No make up tests will be given.**
2. Assignments and research projects/report works will be given throughout the semester. Such requirements shall be due as announced in class. Late submission shall be penalized with grade deductions (5% per day) or shall no longer be accepted, depending on the subject facilitator's discretion. Assignments and exercises are designed to assist you in understanding the materials presented in class, and to prepare you for the exams.
3. Students are required to attend classes regularly, including possible make-up classes. **It is the student's responsibility to gather information missed due to absence.** The university guidelines on attendance and tardiness will be implemented.
4. **Any evidence of copying or cheating during any examinations may result in a failing grade from the examination for all parties involved. Note that other university guidelines shall be used in dealing with this matter.**
5. Students are advised to keep graded work until the semester has ended.
6. Contents of the syllabus are subject to modification with notification.
7. Mobile phones, radios or other listening devices are not allowed to be used inside lecture and laboratory rooms to prevent any distractive interruption of the class activity.
8. No foods, drinks, cigarettes nor children are allowed inside the lecture and laboratory rooms.
9. Withdrawal and dropping from the subject should be done in accordance with existing university policies and guidelines regarding the matter.

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Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

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CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will



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	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Quizzes		Quizzes	
Long Test		Long Test	
Recitation and Attendance		Recitation and Attendance Research	
Research Work		Work	
Assignment	30%	Assignment	30%
Projects		Projects	100%
Exercises		Exercises	
Midterm Examinations		Final Examinations (Teaching	
	100%	Demonstration)	

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**WORLD UNIVERSITY RANKINGS**

2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

**WURI**
The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/PRODUCTION

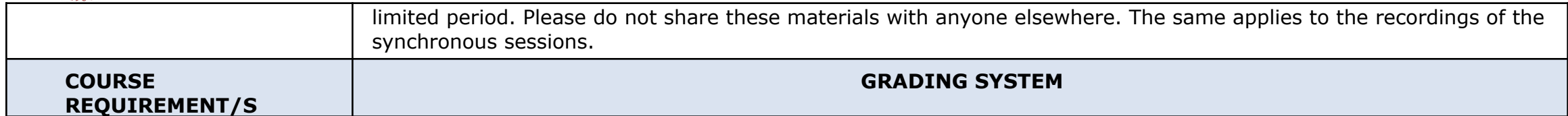
**STARS**


Employability
Online Learning


Teaching
Academic Development
Inclusiveness


Social Responsibility
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

**Times Higher Education Impact Rankings 2024**
DEBUTS AT 1501+

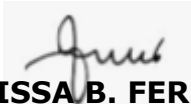


- Making a Learning Plan in Teaching Prose (focusing on Completeness, Appropriate Approach Used, and Facilitative Process) - Teaching Demonstration in Prose or poetry (focusing on Teacher- Like Simulation, Preparation, Classroom Management, Learning Environment, and Interest and Diligence in Teaching Poetry)	FINAL GRADE = (FIRST GRADING + SECOND GRADING) / 2	
CLASS INFORMATION	FACULTY INFORMATION	
Course, Year and Section:	Name of Faculty: MODESTO, LADY MELINDA MINETTE L.	
Day and Time:	Consultation Time:	
Room:	Office Tel. No./ Mobile Phone No.	
School Year: Semester:	E-Mail Address:	



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**QS STARS**


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